

S3KG

1. Triple-Level Matching - each triple in KG_1 is serialised as text ("subject predicate object") and encoded with SBERT. The most similar triple in KG_2 is found by cosine similarity, producing a filtered KG_2 of equal size of kg_1 . This stops the WL kernel from comparing unequal graphs.

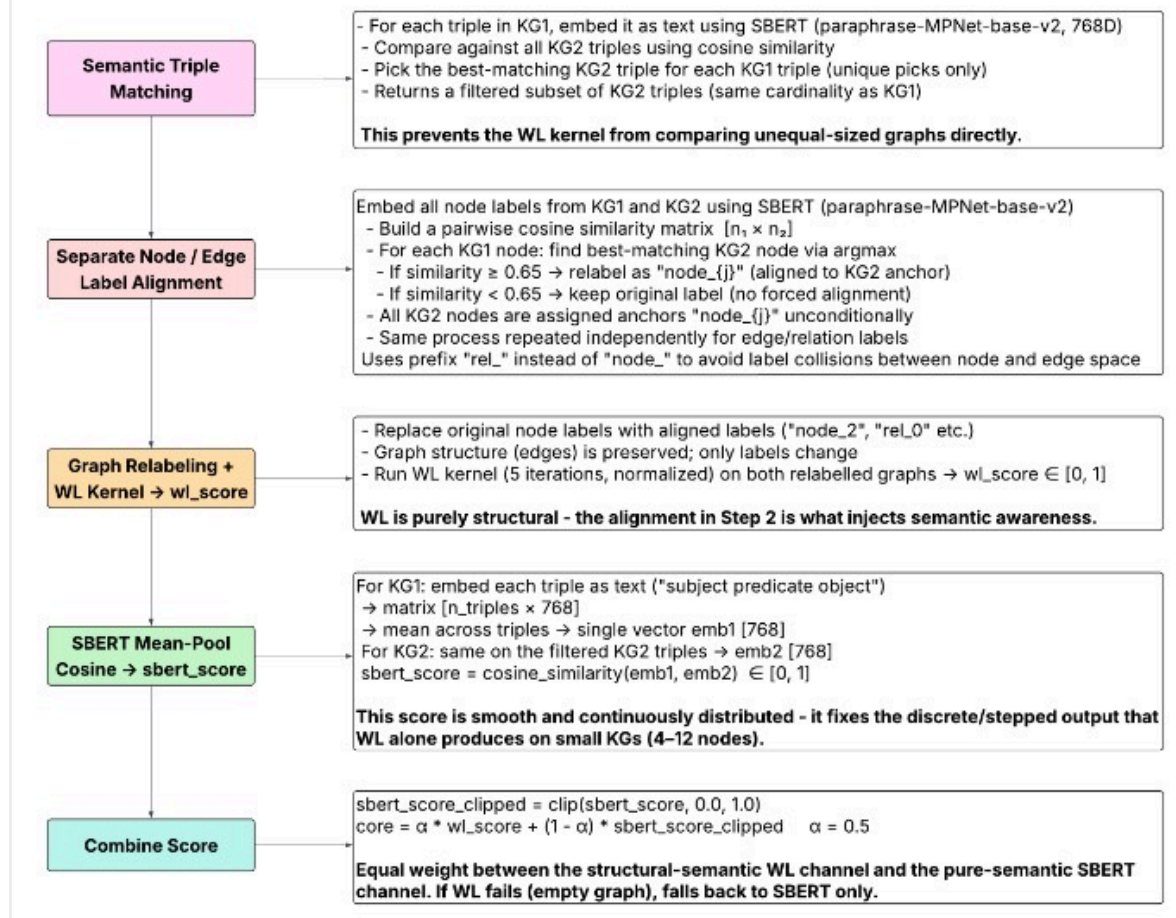
2. Soft Label Alignment - node and relation labels that are almost similar, are aligned by replacing the KG_1 label with the KG_2 if their SBERT cosine similarity is greater than 0.65.

3. WL Kernel (S_{WL}) - runs 5 iterations of neighbourhood label aggregation (like getting nearby entities embeddings and update with 5 rounds) on the aligned graphs (means the same labeled graphs) to produce a structural similarity score.

4. SBERT Mean-Pool (S_{SBERT}) - all triples per KG (overall graph wise) are embedded and mean-pooled into a single 768-D vector and cosine similarity gives a semantic similarity score.

5. Final Score - $S_{S3KG} = \alpha \cdot S_{WL} + (1-\alpha) \cdot S_{SBERT}$, with $\alpha=0.5$ Avg of these 2 structural (triplet wise) and semantical (overall graph wise) similarity scores .

S3KG: Semantic Structural Similarity



LLM Evaluation

For each QA sample, three KGs: from the LLM's answer, the gold answer, and the context. S3KG then computes:

- GoldSim - LLM KG vs gold KG $\rightarrow$ factual accuracy
- CtxSim - LLM KG vs context KG $\rightarrow$ contextual faithfulness
- CUS - harmonic mean of both, penalising imbalance (like f1 score)

400 samples were run per model dataset pair (MesaQA and PubMedQA).

- Mistral-7B leads CUS (0.678 / 0.592) from the table $\rightarrow$ Best at balancing factual accuracy and context faithfulness - that means overall mistral understand the context well and give the answer aligned with the ground truth answer - so this is our best llm here in the context of contextual understanding
- CtxSim > GoldSim across all 8 pairs $\rightarrow$ Models elaborate on context rather than mirror the reference answer concisely - that means here looking at the table ctx similarity is high in 8 cases, this tells models

understand the context well than giving the better answer for the Question.

- ~10 pp CUS gap MesaQA vs PubMedQA --> Biomedical vocabulary and reasoning are the main bottleneck, not model size - this means comparing mesa and pubmed, there are approxly 10% similarity difference. That shows mesa is general health questions while pubmed is little typical medical domain questions, so this shows model struggles when deeper domain specific questions are asked.
- Falcon-7B weakest across all metrics --> Least capable at both factual grounding and context faithfulness - looking at the table in all scores it performs least.